

SKILL-3

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1. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality. To Allocate Buffers Dynamically, Resize Arrays, Prevent BufferOverflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#define MAX 10
int main()
{
    char *history[MAX];
    char command[100];
    int count = 0;
    while (1)
    {
        printf("Sharanya$> ");
        fgets(command, sizeof(command), stdin);
        command[strspn(command, "\n")] = '\0';
        if (strcmp(command, "exit") == 0)
            break;
        if (strcmp(command, "history") == 0)
        {
            for (int i = 0; i < count; i++)
                printf("%d; %s\n", i + 1, history[i]);
            continue;
        }
        if (count < MAX)
        {
            history[count] = malloc(strlen(command) + 1);
            strcpy(history[count], command);
            count++;
        }
        for (int i = 0; i < count; i++)
            free(history[i]);
        return 0;
    }
}
```

Output:

```
sharanaya@DESKTOP-TPK9V3Q:~$ nano skill3a.c
sharanaya@DESKTOP-TPK9V3Q:~$ gcc skill3a.c -o skill3a
sharanaya@DESKTOP-TPK9V3Q:~$ ./skill3a
Sharanya$> ls
Sharanya$> pwd
Sharanya$> history
Sharanya$> whoami
Sharanya$> history
Sharanya$> history
Sharanya$> ls
Sharanya$> pwd
```

2. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output. To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
#include <stdio.h>
#include <string.h>
int main()
{
    char input[100];
    char *token;
    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';
    if (strlen(input) == 0)
    {
        printf("Empty command\n");
        return 0;
    }
    token = strtok(input, " ");
    while (token != NULL)
    {
        printf("Token: %s\n", token);
        token = strtok(NULL, " ");
    }
    return 0;
}
```

Output:

```
sharanaya@DESKTOP-TPK9V3Q:~$ nano skill3b.c
sharanaya@DESKTOP-TPK9V3Q:~$ gcc skill3b.c -o skill3b
sharanaya@DESKTOP-TPK9V3Q:~$ ./skill3b
Enter command: ls 3
Token: ls
Token: 3
sharanaya@DESKTOP-TPK9V3Q:~$ |
```